

Same Facts, Different Interfaces: Evaluating FAQ, Retrieval Chatbot, and AI Chatbot for Student Services

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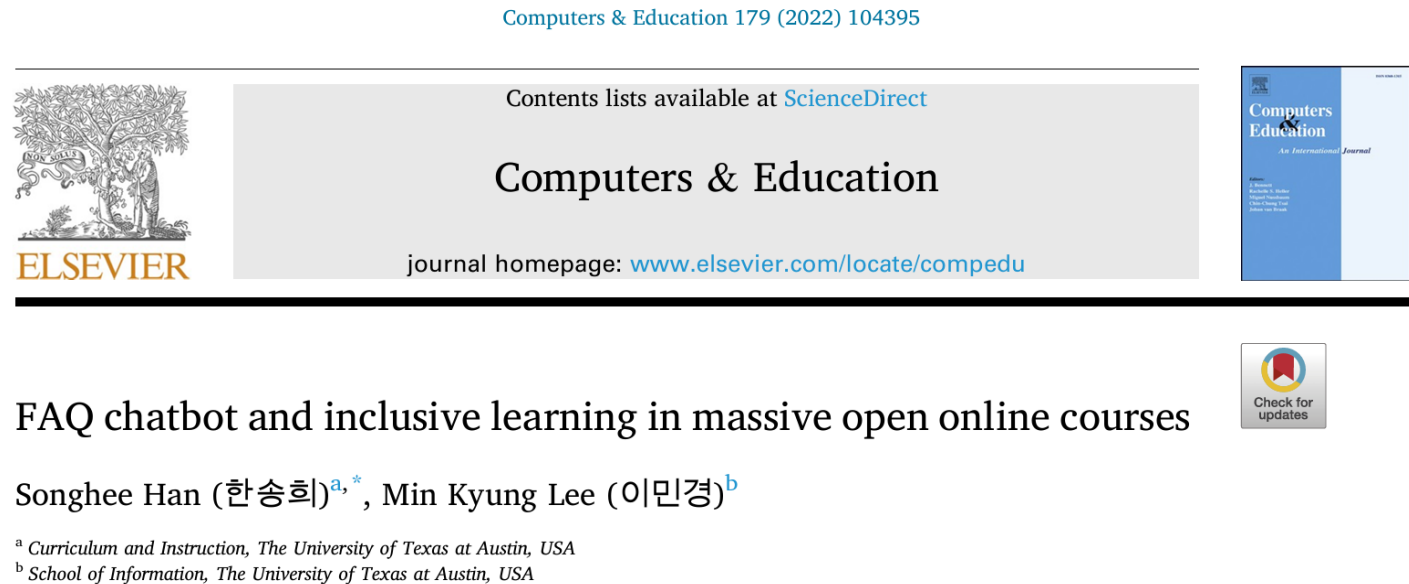
Background

- Universities increasingly use digital tools to answer student questions
- FAQ pages remain a common baseline
- Chatbots are increasingly used in education and student support
- AI chatbots are emerging as a more conversational alternative



Prior Work and Gap

- Prior Work (e.g., Han & Lee, 2022)
 - Compared FAQ webpage and FAQ chatbot using similar content
 - Showed that interface influences user experience (MOOC context)
- Gap
 - Limited to only two interface types
 - Does not include AI chatbot systems
 - Focused on course content rather than student services



Goal & Objectives

Goal

- To evaluate how different information delivery interfaces affect student experience when using the same content

Objectives

- Compare FAQ, retrieval chatbot, and AI chatbot systems
- Measure task completion time and accuracy
- Assess ease of use, mental effort, and trust
- Examine system response speed and cost

Methodology

- Developed three systems:
 - FAQ webpage
 - Retrieval-based chatbot
 - AI-powered chatbot
- Built using a shared knowledge base from a hypothetical university setting
- Dataset designed to reflect real student service information

Choose How You Want To Search

Select the experience that matches the task. You can browse the FAQ directly, use retrieval-based answers grounded in the FAQ database, or open the AI assistant for a more conversational response.

FAQ

Browse questions and answers directly, with category filtering and search.

Best when you want to inspect exact stored answers.

Open FAQ

Retrieval

Ask a question and get the closest answer pulled from the student support knowledge base.

Best when you want fast grounded matching against existing FAQs.

Open Retrieval Chat

AI

Use the conversational assistant for natural responses backed by your existing student support information.

Best when you want a more guided, assistant-style interaction.

Open AI Chat

Methodology

- Study Procedure
 - Participants interacted with all three systems independently
 - Asked the same questions across systems

Evaluating Different Student Information Interfaces

This study evaluates how different systems (FAQ, Chatbot, and AI Chatbot) present student-related information.

You will complete tasks using each system and answer short surveys after each section.

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* Indicates required question

What is your highest level of education? *

☐ High School

☐ Undergraduate (Bachelor's)

☐ Master's

☐ PhD

☐ Other

Are you currently a university student? *

☐ Yes

☐ No

Year of graduation (expected) if you are currently a student.

Your answer

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Instructions

In this study, you will assume the role of a student at a university. The systems you will use are based on a **hypothetical university environment**, but the information reflects real student services such as registration, transcripts, housing, and tuition. You will be given a set of questions that represent common student queries. Provide brief answers (a few words or short phrases are sufficient). You may copy and paste answers from the system if needed.

For each system:

- Read each question carefully
- Click the link provided to access the system
- Use the system to find the answer
- Return to this form and enter your response

Important

- Treat each question as a real student query
- Do not use external sources (e.g., Google)
- Try to answer as accurately and quickly as possible
- Each system contains the **same information**, presented differently

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Never submit passwords through Google Forms.

This form was created inside of Binghamton University.
Does this form look suspicious? [Report](#)

Google Forms

Methodology

- Measures
 - Completed a short survey after each system
 - Task completion time and accuracy
 - Ease of use, mental effort, and trust
 - System response speed and cost

Evaluating Different Student Information Interfaces

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* Indicates required question

Systems Comparison

Which system was easiest to use? *

- ☐ FAQ Webpage
- ☐ Retrieval Chatbot
- ☐ AI Chatbot

Which system did you trust the most? *

- ☐ FAQ Webpage
- ☐ Retrieval Chatbot
- ☐ AI Chatbot

Which system helped you find answers fastest? *

- ☐ FAQ Webpage
- ☐ Retrieval Chatbot
- ☐ AI Chatbot

Which system did you prefer overall and why? *

- ☐ FAQ Webpage
- ☐ Retrieval Chatbot
- ☐ AI Chatbot
- ☐ Other: _____

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Discussion: Systems Evaluation

- What is working well?
- Accurate Answers for known questions

FAQ & Chat Application

Interfaces FAQ Retrieval Chat AI Chat Evaluation

Frequently Asked Questions

Search FAQs Search

All Academic Records and Transcripts Admissions Financial Aid Housing Registration and Enrollment Student Services Tuition and Billing

Admissions

What are the undergraduate admission requirements for first-year applicants? +

What steps are required to complete an undergraduate application for admission? +

What are the application deadlines for fall and spring admission terms? +

Is an application fee required for admission, and are fee waivers available? +

How can applicants monitor the status of their submitted application materials? +

FAQ & Chat Application

Interfaces FAQ Retrieval Chat AI Chat Evaluation

FAQ Assistant

Ask me about admissions, financial aid, housing, registration, transcripts, and other student support topics.

Hello! I'm here to help with questions across admissions, financial aid, housing, registration, transcripts, and student services. What would you like to know?

Type your question here... Send

FAQ & Chat Application

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AI FAQ Assistant

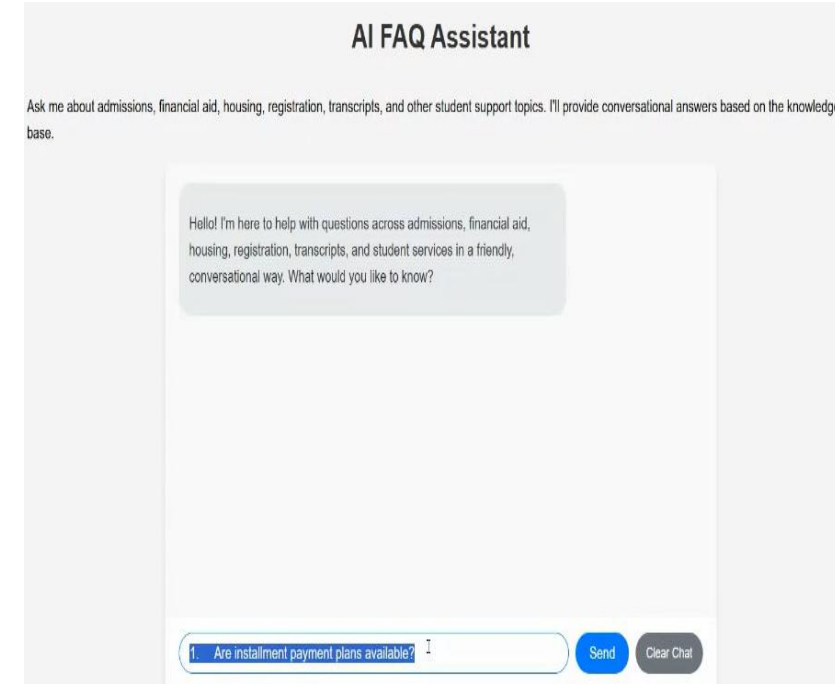
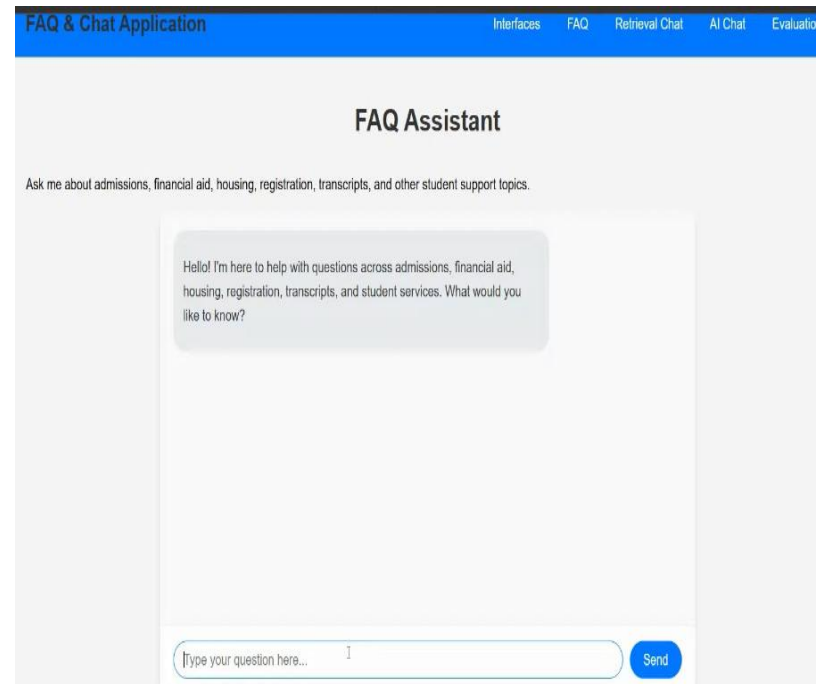
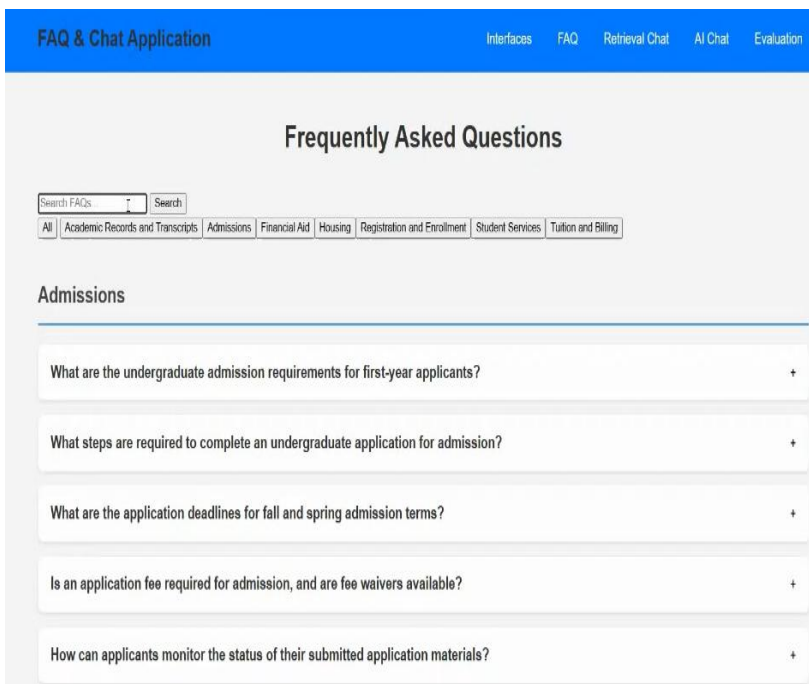
Ask me about admissions, financial aid, housing, registration, transcripts, and other student support topics. I'll provide conversational answers based on the knowledge base.

Hello! I'm here to help with questions across admissions, financial aid, housing, registration, transcripts, and student services in a friendly, conversational way. What would you like to know?

Type your question here... Send Clear Chat

Discussion: System Evaluation (Limitations)

- FAQ
 - Long list of Questions
 - Cannot answer paraphrased questions if keywords do not exist
- Retrieval Chatbot
 - Cannot answer paraphrased questions if keywords do not exist
- AI Chatbot
 - T



Discussion: Participants' Experience

Discussion: Participants' Experience

FAQ & Chat Application

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Frequently Asked Questions

What are the official course

Search

All

Academic Records and Transcripts

Admissions

Financial Aid

Housing

Registration and Enrollment

Student Services

Tuition and Billing

Registration and Enrollment

What are the official course registration periods for each academic term?

-

Course registration typically takes place in April for the First Semester and November for the Second Semester.

Source: FAQ #11 - Registration and Enrollment

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FAQ Assistant

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FAQ & Chat Application

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AI FAQ Assistant

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Conclusions & Future Work

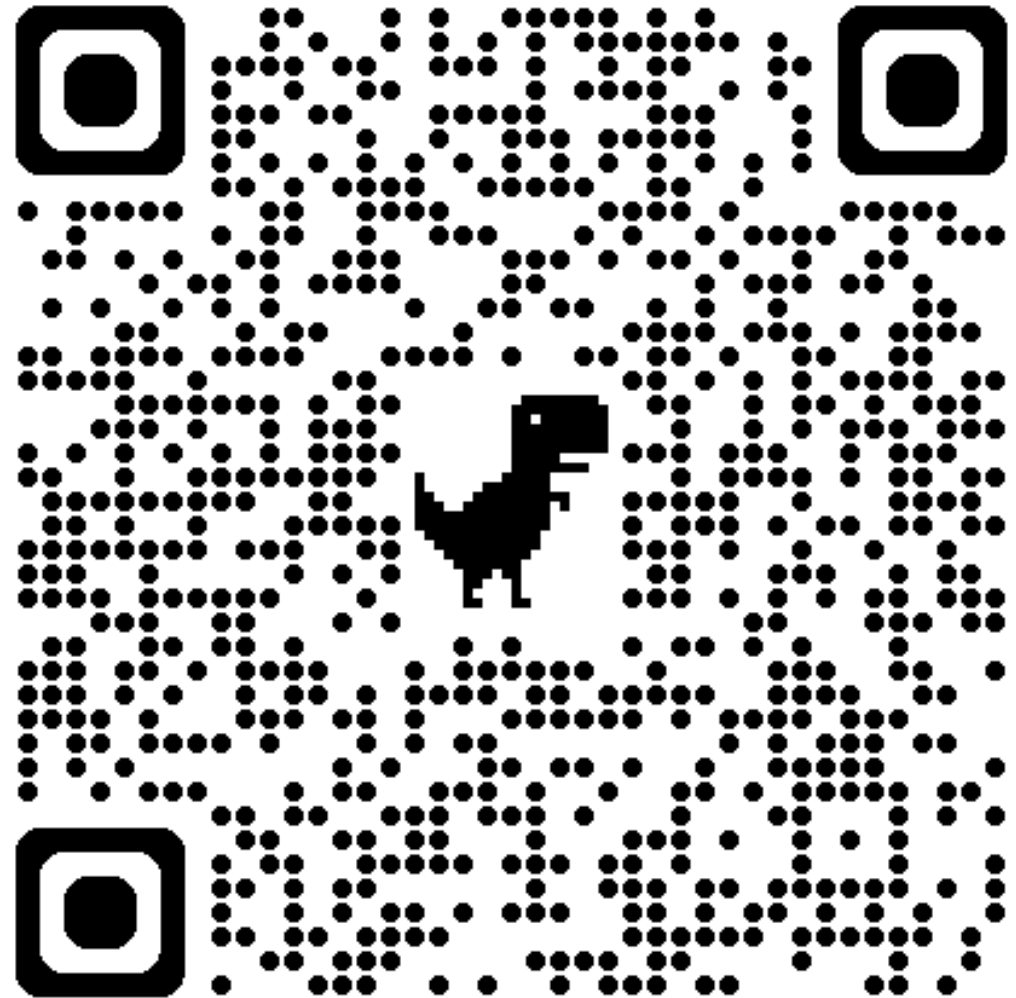
- Preliminary Results
 - Participants enjoyed using Retrieval Chatbot and AI Chatbot the most
 - AI Chatbot was too chatty for some participants
 - System evaluation showed that AI Chatbot behaved the best compared to the two other systems
- Future Work
 - Fine tune the built systems and making it active while offline
 - Get at least 50 participants for more robust results
 - Funding opportunities to compensate participants

References

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Thank you!!!

- Scan the QR Code to participate
- Contact me at hkddakpo@st.ug.edu.gh for collaboration



Please Scan Me